PROPERTIES OF MORPHISMS UNDER THE NEAR-POINT GLUEING EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. Author delegation kept verbatim for part (2): 'Thus we can use exactly the same arguments as in the proof of (1).' The argument for (1) is fully present in span and the adaptation to (2) is the system-level bookkeeping; flagged for human pass.

Allowed standard prerequisites (editorial).

- Near-point glueing. Limits, Lemma lemma-glueing-near-point (same chapter)
- Descent engines. Limits, descend lemmas of the chapter

Source reference. Original lemma 0BQ5: sources/stacks-chow/tags.txt, source lines 14723–14723. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZQ: sources/stacks-chow/tags.txt, source lines 2603–2603. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZO: sources/stacks-chow/tags.txt, source lines 2601–2601. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 07RP: sources/stacks-chow/tags.txt, source lines 9874–9874. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Notation and assumptions as in Lemma 0BQ5. Let $U \subset U' \subset X$ be an open containing s.*

(1) *Let $f' : X \to U'$ correspond to $f : X' \to U$ and $g : Y \to \mathrm{Spec}(\mathcal{O}_{S,s})$ via the equivalence. If f and g are separated, proper, finite, étale, then after possibly shrinking U' the morphism f' has the same property.*

(2) *Let $a : X_1 \to X_2$ be a morphism of schemes of finite presentation over U' with base change $a' : X_1' \to X_2'$ over U and $b : Y_1 \to Y_2$ over $\mathrm{Spec}(\mathcal{O}_{S,s})$. If a' and b are separated, proper, finite, étale, then after possibly shrinking U' the morphism a has the same property.*

Proof. Proof of (1). Recall that $\mathrm{Spec}(\mathcal{O}_{S,s})$ is the limit of the affine open neighbourhoods of s in S. Since g has the property in question, then the restriction of f' to one of these affine open neighbourhoods does too, see Lemmas 01ZQ, 081F, 01ZO, and 07RP. Since f' has the given property over U as f does, we conclude as one can check the property locally on the base.

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Proof of (2). If we write $\mathrm{Spec}(\mathcal{O}_{S,s}) = \lim W$ where W runs over the affine open neighbourhoods of s in S, then we have $Y_i = \lim W \times_S X_i$. Thus we can use exactly the same arguments as in the proof of (1). $\square$